STACKS, MODULI OF CURVES: LEMMA-CONTRACT-BASIC-UNIQUENESS

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 2899-2912 of sources/stacks-moduli-curves/pinned-original-source.tex; proof span is lines 2914-2984, the complete author proof (69 body lines).

Editorial note. Navigation stubs: 8 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 4 of queue_moduli-curves.json (pb=3316, stmt 2899-2912, proof 2914-2984).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 0A42: sources/stacks-moduli-curves/pinned-original-source.tex, source lines 2984–2984. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0E3X: sources/stacks-moduli-curves/pinned-original-source.tex, source lines 2984–2984. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0DSS: sources/stacks-moduli-curves/pinned-original-source.tex, source lines 655–658. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 07NU: sources/stacks-moduli-curves/pinned-original-source.tex, source lines 2984–2984. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 07SK: sources/stacks-moduli-curves/pinned-original-source.tex, source lines 2984–2984. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0AE7: sources/stacks-moduli-curves/pinned-original-source.tex, source lines 2984–2984. Cited by the retained statement or

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proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 089E: sources/stacks-moduli-curves/pinned-original-source.tex, source lines 2984–2984. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0CMV: sources/stacks-moduli-curves/pinned-original-source.tex, source lines 2984–2984. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let S be a scheme and $s \in S$ a point. Let $f : X \to S$ and $g_i : Y_i \to S$, $i = 1, 2$ be families of curves. Let $c_i : X \to Y_i$ be morphisms over S. Assume there is an isomorphism $Y_{1,s} \cong Y_{2,s}$ of fibres compatible with $c_{1,s}$ and $c_{2,s}$. If $c_{1,s,*}\mathcal{O}_{X_s} = \mathcal{O}_{Y_{1,s}}$ and $R^1c_{1,s,*}\mathcal{O}_{X_s} = 0$, then there exist an open neighbourhood U of s and an isomorphism $Y_{1,U} \cong Y_{2,U}$ of families of curves over U compatible with the given isomorphism of fibres and with c_1 and c_2.*

Proof. Recall that $\mathcal{O}_{S,s} = \operatorname{colim} \mathcal{O}_S(U)$ where the colimit is over the system of affine neighbourhoods U of s. Thus the category of algebraic spaces of finite presentation over the local ring is the colimit of the categories of algebraic spaces of finite presentation over the affine neighbourhoods of s. See Limits of Spaces, Lemma 07SK. In this way we reduce to the case where S is the spectrum of a local ring and s is the closed point.

Assume $S = \operatorname{Spec}(A)$ where A is a local ring and s is the closed point. Write $A = \operatorname{colim} A_j$ with A_j local Noetherian (say essentially of finite type over $\mathbf{Z}$) and local transition homomorphisms. Set $S_j = \operatorname{Spec}(A_j)$ with closed point s_j. We can find a j and families of curves $X_j \to S_j$, $Y_{j,i} \to S_j$, see Lemma 0DSS and Limits of Stacks, Lemma 0CMV. After possibly increasing j we can find morphisms $c_{j,i} : X_j \to Y_{j,i}$ whose base change to s is c_i, see Limits of Spaces, Lemma 07SK. Since $\kappa(s) = \operatorname{colim} \kappa(s_j)$ we can similarly assume there is an isomorphism $Y_{j,1,s_j} \cong Y_{j,2,s_j}$ compatible with $c_{j,1,s_j}$ and $c_{j,2,s_j}$. Finally, the assumptions $c_{1,s,*}\mathcal{O}_{X_s} = \mathcal{O}_{Y_{1,s}}$ and $R^1c_{1,s,*}\mathcal{O}_{X_s} = 0$ are inherited by $c_{j,1,s_j}$ because $\{s_j \to s\}$ is an fpqc covering and $c_{1,s}$ is the base of $c_{j,1,s_j}$ by this covering (details omitted). In this way we reduce the lemma to the case discussed in the next paragraph.

Assume S is the spectrum of a Noetherian local ring Λ and s is the closed point. Consider the scheme theoretic image Z of

$$(c_1, c_2) : X \longrightarrow Y_1 \times_S Y_2$$

The statement of the lemma is equivalent to the assertion that Z maps isomorphically to Y_1 and Y_2 via the projection morphisms. Since taking the scheme theoretic image of this morphism commutes with flat base change (Morphisms of Spaces, Lemma 089E, we may replace Λ by its completion (More on Algebra, Section 07NU).

Assume S is the spectrum of a complete Noetherian local ring Λ. Observe that X, Y_1, Y_2 are schemes in this case (More on Morphisms of Spaces, Lemma 0AE7). Denote X_n, $Y_{1,n}$, $Y_{2,n}$ the base changes of X, Y_1, Y_2 to $\operatorname{Spec}(\Lambda/\mathfrak{m}^{n+1})$. Recall that the arrow

$$\mathcal{D}ef_{X_s \to Y_{2,s}} \cong \mathcal{D}ef_{X_s \to Y_{1,s}} \longrightarrow \mathcal{D}ef_{X_s}$$

is an equivalence, see Deformation Problems, Lemma 0E3X. Thus there is an isomorphism of formal objects $(X_n \to Y_{1,n}) \cong (X_n \to Y_{2,n})$ of $\mathcal{D}ef_{X_s \to Y_{1,s}}$. Finally, by Grothendieck's algebraization theorem (Cohomology of Schemes, Lemma 0A42) this produces an isomorphism $Y_1 \to Y_2$ compatible with c_1 and c_2. □